

1. Executive Summary

This is a self-built, autonomous PAPER-TRADING agent for Indian index option chains (NIFTY 50, BANKNIFTY; SENSEX/FINNIFTY supported by design). It runs 09:15–15:30 IST Mon–Fri against LIVE market data (FYERS WebSocket primary, Yahoo Finance fallback), registers real option contracts (expiry, strike, CE/PE, lot size), and opens paper positions ONLY on option contracts — never on the raw index. Capital is a ■5,000 paper envelope that grows/shrinks by realized (profit – charges).

The agent is at an early, honest stage: the DATA + EXECUTION + DECISION plumbing is real and verified end-to-end, but the SIGNAL layer is still a heuristic (SMA-20 direction + ATM/strike scoring), there is no ML yet, no backtesting, and exactly two paper trades have ever been closed (both legacy index-level trades from an earlier scope-drift, closed deliberately). This report is written for external review: it states what exists, which logic was chosen and WHY, where design conflicts occurred, and the concrete gaps that block the path to a successful option-chain trader.

2. What Is Implemented (verified live)

2.1 Runtime topology

Layer	Detail
Host	Dhargent VM 168.110.60.10 — headless NestJS service "trading-agent" (pm2), no git; code lives in the dev repo
Dev repo	my-job-agent (NestJS + TypeORM + MySQL on OCI MDS); commits on branch dev, pushed origin/dev
Market data	FYERS v3 WebSocket (primary) + Yahoo Finance fallback when FYERS token/feed is down (documented, sanctioned)
Subscription	3 index underlyings + 32 option contracts = 35 FYERS symbols streamed every tick
DB (shared)	fnf_portfolios, fnf_trades, fnf_option_contracts, fnf_option_quotes, fnf_market_snapshots (+ *_history archive tables)

2.2 Data engine & archival

Every tick is stored: index ticks → fnf_market_snapshots (today table); option ticks (premium, bid/ask, volume, OI, Greeks when FYERS sends them) → fnf_option_quotes. Session-scoped archival: at 15:30 close and at IST day rollover, ticks move INSERT...SELECT into fnf_market_snapshots_history / fnf_option_quotes_history and are deleted from the live tables — so live tables always hold only the current session/day, and every historical tick is preserved (35,965 snapshot rows archived to date). Contract symbols in FNO_MARKET_DATA_SYMBOLS auto-register into fnf_option_contracts on boot.

2.3 Registered option universe (as of report date)

Underlying	Spot (last)	Expiry series	Registered	Lot size
NIFTY50-INDEX	23,897.70	26SEP 2026	24 contracts (8×CE+PE band)	65
NIFTYBANK-INDEX	57,369.65	26SEP 2026	16 contracts	30
SENSEX / FINNIFTY	76,515 / 26,051	—	not yet (quotes probed live)	20 / 60

2.4 Execution engine (session driver)

Session driver ticks every 10s inside the IST window. It (a) manages EXITS of every open position against its own live price source and the target/stop stored at entry (position-driven, not signal-driven — legacy index positions remain manageable), and (b) consumes the decision engine's output to OPEN at most one new position per eligible signal. Costs use a FYERS per-segment model (options flat ■20/order, STT 0.05% sell side, NSE txn, stamp, SEBI, GST 18%); brokerage min/max bug fixed. P&L; is premium-based: (exit – entry) × units (lots × lotSize).

2.5 Decision engine (option-candidate-rank-v1) — the core logic

Signal flow, per your architecture direction: Direction → Candidate build → Capital filter → Liquidity/risk gates → Score → Single best contract or NO TRADE.

1. DIRECTION: SMA-20 + momentum on each underlying index (bullish → CE side, bearish → PE side; range-bound → no candidates). Decay-aware confidence gate (time-of-day and data-age decay).

2. CANDIDATE BUILD: every registered option contract matching the direction's option type with a live premium becomes a candidate carrying underlying, expiry, strike, CE/PE, premium, lot size,

contract value, bid/ask spread, volume/OI, Greeks (where quoted), ATM-distance, days-to-expiry.

3. CAPITAL FILTER (■5,000 is a filter, NOT an instrument rule): reject any contract where $\text{premium} \times \text{lot_size} \times \text{lots} > \text{headroom}$, $\text{headroom} = (\text{capital} + \text{netPnl}) - \text{deployed}$.

4. MANDATORY GATES: fresh quote (≤ 5 min), premium > 0 , expiry $\geq$ tomorrow, spread $\leq 15\%$, min volume/OI floor, decayed confidence $\geq$ floor.

5. SCORE (env-tunable weights): strategy fit/ATM-ness 40 (6% band — widened because with ■5k only OTM strikes are affordable and the score must still separate near-OTM from deep-OTM), liquidity 25, spread 20, expiry quality 15 (3–14 DTE preferred), Greeks fit 10, same-underlying continuity +5 (scoring preference only — NEVER a hard fallback), capital-efficiency 5.

6. OUTPUT: the single highest-scoring affordable contract is emitted as BUY. If every candidate fails the mandatory gates $\rightarrow$ NO TRADE (never force the next-cheapest). Verified: NO TRADE after hours (stale gate), NO TRADE when headroom was negative (legacy positions), correct ranking over the 32-contract universe during the live test.

3. Logic Choices and the Reasoning Behind Them

use the index as the underlying/reference, but trade the corresponding option-chain contracts. Never treat NIFTY50-INDEX or SENSEX as "underlying".
 added "NIFTY unaffordable → SENSEX". Corrected: the constraint filters every candidate contract; SENSEX/BANKNIFTY/FINNIFTY and NIFTY are not.
 d max loss = premium; writing options has unbounded risk. Fits the envelope and the paper-risk policy.
 unaffordable. Affordable contracts are 1.5–5% OTM; a 2% band would zero out strategy fit for all survivors and let liquidity alone decide.
 the premium outlay; index-notional accounting was the earlier bug (■81k "deployed" against ■5k).
 a fresh signal for the same instrument each tick; the position knows its own price source and levels.
 n-pure, preserves every tick for later feature/ML work; user requirement (2026-09-04).
 after each close ceiling = $\max(0, \text{capital} + \text{netPnl})$. Losses and service charges reduce the envelope; profits grow it.
 e strategy/liquidity/risk intent — flat is a position.

3.1 Cost model (FYERS, per executed order)

Component	Options	Futures
Brokerage	■20 flat	min(0.03% × turnover, ■20)
STT (sell side)	0.05% of premium	~0.05% of notional
NSE txn charges	0.0355% of premium	0.00183%
Stamp (buy side)	0.003%	0.002%
SEBI + GST 18%	■10/crore + 18% on (brokerage+txn+SEBI)	same

Note: cost rates are from FYERS official pages (researched 2026-09-03); figures marked ~ should be re-verified before any real-money use.

4. Design Conflicts & Decisions Made (history as evidence)

index positions were later closed at user's instruction (net -■200.74 incl. ■64.54 charges) to release ■81k phantom "deployed".

rationalization. User: "my option chain ceiling was 5k not 100k". Corrected to ceiling=capital=■5,000, then ceiling=capital+netPnl.

AMPLE, not a prescribed fallback. Rewrote docs + engine to the capital-filter model; SENSEX is just another candidate.

at T-03 open (error -15). Feed fell back to Yahoo (sanctioned). User provided a fresh auth-code click; token exchanged, primary restore

provided, 2026-09-04) is Data → Features → Regime → AI signal (XGBoost/LightGBM, later DeepLOB) → Decision → Paper → Learning

TC; queries need explicit CONVERT_TZ. Fixed in the archive logic by comparing against IST boundaries in the same naive frame.

5. What the Agent Is Missing to Become a Successful Option-Chain Trader

Honest self-assessment — ordered roughly by criticality.

5.1 Data & market-structure gaps

	Impact
or study	35k index-tick rows exist but option-quote history is only days old — no way yet to study premium behavior
by FYERS	Score uses Greeks only "where available"; delta/theta inputs are often absent → greeksFit is a weak signal
microprice/OFI/GEX features	Feature engine is SMA-only. No volatility context (IV rank/percentile), no breadth, no order-flow.
red (26SEP)	No weekly vs monthly comparison; theta behavior across expiries untested.
ape ingestion	Realistic fill modeling (spread/slippage/latency) and DeepLOB are impossible until this exists.
not registered yet	The "search the universe" rule is only as good as the universe; 2 underlyings today.

5.2 Strategy & risk gaps

	Impact
	No option-aware signal: premium dynamics, IV, theta decay, and event risk are not modeled. Options are traded like leverage
	ATM/OTM directionality only — no spreads, straddles, covered calls, iron condors, or delta-hedged structures; many of the
	Deep OTM options are high-theta, low-probability "lottery tickets" — the plan doc itself warns this is capital-destructive; the
envelope	Envelope protects per-close, but there is no session P&L limit, no consecutive-loss pause, no volatility shutdown.
	All positions are opened intraday-window; what should happen if a position legitimately spans sessions (decay, gap risk) is
	It is a design signature (user belief system) but has no statistical evidence; it may help or hurt — needs A/B or at least logg

5.3 Engineering & validation gaps

	Impact
	No historical replay of the decision engine; every rule (weights, 6% band, gates) is untested against the past.
	The learning loop has no data: no failure analysis, no retrain, no model comparison. XGBoost on zero labels is impossible — the
	Not failure-mode identification; no per-strategy/per-strike/per-expiry attribution.
	The user architecture's promotion pipeline does not exist (by design — paper first).
ily)	Single-day access token; if the session host reboots or the token dies overnight, feed degrades to Yahoo until a human logs in.
	40/25/20/15/10/5/5 were chosen by reasoning, not fitted to any outcome data.
	The agent runs headless; events are logged but there is no push channel for "opened", "stopped out", "token expired".

6. Questions for Reviewers (where external input is most valuable)

1. Given a ₹5,000 envelope and NIFTY near 23,900 (1 lot ≈ ₹13–26k at ATM), what is the most defensible option-buying structure — deep-OTM single contracts, far-OTM with tight stops, smaller-lot chains (BANKNIFTY 30 / FINNIFTY 60 / SENSEX 20), or should the envelope itself be reconsidered?
2. Which option-aware features should the feature engine prioritize first (IV rank, term structure, open-interest delta, put/call ratio, VWAP deviation) given we only have tick OHLC + premium + OI now?
3. What is the minimum realistic paper-trade count before XGBoost/LightGBM becomes meaningful, and what label design would you use (target/stop outcome? time-decayed premium P&L;? theta-adjusted?)
4. Are the score weights and the 6% ATM band sane for a long-options-only desk, or should the strategy term use a different objective (e.g. $\max(\text{prob} \times \text{payoff})$ via delta and expected move)?
5. What is the right overnight/multi-session policy for paper options given the driver only runs 09:15–15:30?
6. Is a single-position-per-signal cap correct for ₹5k, or should the engine be allowed concurrent uncorrelated positions when each fits the envelope?
7. Which mandatory gates would you add (e.g. IV-percentile floor, minimum OI, max spread in absolute rupees, DTE minimum 7)?
8. How should the muhurta/astro window be tested honestly alongside the quant signal without confounding either?

Appendix A — Current Live State (2026-09-04 22:18 IST)

	Value
	sandbox-live — capital █5,000.00, deployed █0.00, netPnl -█200.74, ceiling █4,799.26, totalCost █64.54, auto
des	2 (legacy index BUYs closed at user instruction)
es	0
d contracts	40 (24 NIFTY + 16 BANKNIFTY, 26SEP)
on quotes (last 60m)	40 distinct contracts
bscription	35 symbols (3 index + 32 contracts)
spots	NIFTY 23,897.70 · BANKNIFTY 57,369.65 · SENSEX 76,515.43
host	Dhargent trading-agent pid 30206 (new build), driver armed 09:15–15:30 IST
	11f773e — decision engine + 32-contract universe committed & pushed origin/dev

— End of report —